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Operationalizing EcoNet Governance: A Research Framework for Verifying Ecosafety Invariants Before Orchestration

This report defines a comprehensive research framework for establishing Virta-Sys as the governance-first headquarters of the EcoNet constellation. The central objective is to create a robust, scalable, and safe architectural system that prioritizes the enforcement of a frozen ecosafety contract over energy-cost optimization or complex orchestration. This framework operationalizes core principles—such as Lyapunov-based risk models, lane-based deployment stages, and cross-band alignment—by transforming them into verifiable, machine-readable artifacts like database schemas, ALN shards, and Rust traits. The analysis systematically deconstructs the proposed architecture, validating its conceptual soundness against established principles from control theory, software engineering, and AI safety. It identifies critical gaps and proposes concrete upgrade pathways, including new Rust traits, SQL schema extensions, and ALN shard specifications, to move the system from a high-level design to a tangible, verifiable platform. The resulting structure positions Virta-Sys not as a simple orchestrator but as the authoritative, non-actuating decision engine that ensures all actions within the constellation remain within a predefined, ecologically safe operating envelope.

The Foundational Pillar: A Frozen Ecosafety Grammar and Spine Architecture

The cornerstone of the EcoNet constellation architecture is the concept of a "frozen" or immutable ecosafety grammar. This foundational pillar transforms abstract environmental stewardship into a verifiable, computationally enforced contract, which is the primary mechanism for mitigating risks in a large-scale, distributed development environment [1] [3]. The architecture rests on three core components: a Lyapunov-based risk model, a set of dedicated "spine" repositories that serve as the single source of truth, and a strict spine/role-band governance model that enforces separation of concerns. Together, these elements create a system where safety is a structural property, not an afterthought, enabling predictable and auditable behavior at scale. The mathematical

bedrock of this grammar is the Lyapunov residual, defined as $V_t = \sum_j w_j r_j^2$, which serves as a global measure of ecological health by aggregating risk coordinates across multiple ecological planes [10] [18]. Each coordinate, r_j, corresponds to a distinct plane such as energy, hydraulics, biology, carbon, materials, biodiversity, and data quality, and is clamped to the interval $[0,1]$ [36]. The weights, w_j, reflect the relative importance of each plane, ensuring the residual is a convex and non-negative quantity [6]. Central to this model is the "safestep" invariant, which mandates that any transition from time t to $t+1$ must satisfy the condition $V_{t+1} \leq V_t$ [91]. This powerful constraint makes ecologically degrading trajectories structurally non-deployable, as any action causing the system's overall risk to increase is automatically rejected by the safety gatekeeper [35].

From this residual, the Knowledge-Eco-Risk (KER) score is derived, providing a standardized way to evaluate performance. Risk is defined as the maximum single-plane risk, $R = \max(r_j)$, while Eco-Impact is its complement, $E = 1 - R$ [36]. Production deployments are subject to strict gates requiring a Knowledge factor (K) greater than or equal to 0.90, an Eco-Impact factor (E) greater than or equal to 0.90, and a Risk factor (R) less than or equal to 0.13 [64] [68]. This formalized mathematics provides the ultimate safety guarantee, turning environmental stewardship into a verifiable computational problem. To enforce this frozen grammar across the entire constellation, a dedicated set of "spine" repositories has been defined: `EcoNet`, `aln-platform-ecosystem`, `Eco-Fort`, and `cydroid-toolkit` [92] [148]. These repositories form the canonical home for all shared artifacts, preventing semantic fragmentation and ensuring consistency [117][126]. They house the core ecosafety logic, ALN specifications, validation tools, and the ALN-driven toolchain for generating cross-language schemas [99] [117]. All other repositories within the constellation are mandated to depend on these four spine repositories for their ecosafety grammar and ALN schemas, rather than inventing their own [117]. This dependency structure is the primary mechanism for enforcing the frozen grammar, ensuring that every piece of code and data shard adheres to the same vetted, high-knowledge standards. This prevents the emergence of conflicting interpretations of "risk" or "data quality" and maintains a coherent semantic layer across the entire system [116][117].

The final component of this foundational pillar is the spine/role-band architecture, which imposes a strict separation of concerns on the repositories themselves. Each repository is assigned to exactly one role band—SPINE, RESEARCH, ENGINE, MATERIAL, GOV, or APP—which dictates its responsibilities, capabilities, and constraints [16] [49]. This clean division of labor is a critical enabler of safety and scalability. For instance, a `RESEARCH` band repository is strictly non-actuating; its sole purpose is to generate evidence, analyze data, and tighten corridors without ever owning or controlling physical actuators [33] [44].

An `ENGINE` band repository, on the other hand, is permitted to implement controllers and physical-domain kernels, but it remains subordinate to the ecosafety spine, which retains ultimate authority over deployment decisions [47] [48]. Similarly, a `MATERIAL` band repo focuses on biodegradable substrates and species corridors, while a `GOV` repo handles financial infrastructures, routing, and governance using shard evidence and DIDs [46]. By assigning a single, well-defined role to each repository, the architecture simplifies governance, streamlines CI/CD pipelines, and ensures consistent KER scoring. It creates clear boundaries that prevent a repository from inadvertently stepping outside its designated domain, thus reducing the potential for misconfiguration and misuse. The following table outlines the role bands and their corresponding responsibilities.

Role Band	Primary Responsibility	Allowed Programming Layers	Key Examples
SPINE	Host the frozen ecosafety grammar, core KER math, and ALN schema library [92].	CORE GRAMMAR	`EcoNet`, `aln-platform-ecosystem`, `Eco-Fort`, `cydroid-toolkit` [148]
RESEARCH	Generate non-actuating workloads, KER scores, corridor updates, and evidence shards [101].	NON-ACTUATING ANALYSIS	`eco_restoration_shard`, `SnowGlobe`, `EcoNet-CERG`
ENGINE	Implement physical-domain controllers and kernels under the ecosafety spine [47].	KERNEL, CONTROLLER, EDGE_SCRIPT	`EcoNet-CEIM-PhoenixWater`, `EcoNetPhoenix`, `Sewer-FOG-Monitoring-Network`
MATERIAL	Develop biodegradable substrates, material kinetics, and species corridor definitions [148].	MATERIAL_MODEL, GOVERNANCE_INTERFACE	`BugsLife`, `Ant-One-Net`, `EcoNet-BeeSafeAI`
GOV	Manage infrastructure governance, workload orchestration, financial flows, and rights [46].	LEDGER, ROUTER, IDENTITY	`eco_infra-governance`, `ecological-orchestrator`, `Paycomp`, `Ocu-Trust`
APP	Provide specialized applications, dashboards, frontends, or user-facing tools.	UI, CONFIG_TOOL	`EcoNetCybocinderPhoenix`, `Eco_Build`, `Windminer`

In summary, the foundation of the EcoNet constellation is a tightly integrated system of mathematical rigor, centralized control, and strict governance. The frozen ecosafety grammar provides the unchangeable rules of the game. The spine repositories act as the constitution, providing the official implementations of those rules. Finally, the role-band architecture enforces discipline, creating a predictable and auditable environment where safety is a structural property, not an afterthought. This design directly addresses the challenge of managing complexity in a distributed system focused on a high-stakes goal like ecological restoration, ensuring that all participants operate from the same playbook.

The Discovery Spine: A Machine-Readable Manifesto for Autonomous Agents

The creation of a SQLite-based discovery spine is the pivotal step in operationalizing the EcoNet architecture, serving as the operational heart of the constellation. This component is far more than a simple file locator; it is a dynamic, queryable manifest that translates the high-level governance rules of the spine architecture into a low-level, actionable intelligence layer for autonomous agents [42] [57]. By providing a standardized vocabulary of roles, shard types, risk planes, and allowed programming layers, it eliminates guesswork and heuristic-based discovery, making reliable, automated cross-repository code generation and workload orchestration possible [140][141]. The design formalizes the existing ALN schemas, corridor definitions, and shard manifests, which were previously documented only in prose or scattered across various files, into a cohesive database schema [117][148]. This formalization is crucial for scalability, as manual oversight becomes untenable as the number of repositories grows. The proposed SQLite schema is comprehensive and meticulously designed to index all relevant entities within the constellation. Its tables provide a multi-faceted view of the system's state.

The `repo` table serves as the top-level catalog of all repositories in the constellation, storing essential metadata such as its name, GitHub slug, visibility, primary language, and its assigned `role_band` [117]. This table is the anchor point for all other queries. The `repo_file` table provides a granular index of files, linking each file to a specific repository via a foreign key. It captures the file's relative path, filename, extension, and classifies it into categories like `QPUDATASHARD`, `PARTICLE`, `SCHEMA`, or `SRC` based on its location in the filesystem [125]. This allows agents to quickly filter for "interesting" ALN or CSV files across the entire constellation. Building upon the file index, the `aln_schema` and `aln_particle` tables create a registry of the ALN grammar itself. The `aln_schema` table maps each indexed file to its logical schema, recording the schema's name (e.g., `ecosafety.riskvector.v2`), version, and category (e.g., `ECOSAFETY`, `FLOWVAC`) [29]. The `aln_particle` table then breaks down these schemas into individual particle definitions, indexing their name (e.g., `HydrologicalBufferPhoenix2026v1`), role (e.g., `QPUDATASHARD`, `SUBSTRATE`), and the Lyapunov channel they belong to (e.g., `hydraulics`, `materials`) [148]. This hierarchical indexing allows an agent to understand not just *what* a file is, but *what it means* within the broader context of the ecosafety planes. The `aln_field` table provides the finest level of detail, indexing every field within a particle definition, noting its name (e.g., `rcarbon`, `rbiodiversity`), data type, units, and whether it is a risk coordinate or part of the KER calculation [95].

To make the discovery process truly powerful, the schema also includes tables for indexing specific shard instances and their properties. The `corridor_definition` table mirrors the `ecosafety.corridors.v2` ALN shard, storing the mandatory safety bands (`safe`, `gold`, `hard`), weights, and Lyapunov channels for each risk coordinate (`varid`). This allows an agent to programmatically discover the safety constraints applicable to a given piece of data [206]. The `shard_instance` table acts as a registry of concrete evidence shards, capturing high-level attributes such as the node ID, region, time window (`t_start_utc`, `t_end_utc`), and the lane (`RESEARCH`, `PROD`) in which the shard was generated [117]. Crucially, it also stores the computed KER metrics (`k_metric`, `e_metric`, `r_metric`) and the `kerdeployable` flag for each shard instance. This enables sophisticated queries like, "Find all hydrological buffer shards in the Phoenix region produced in the PROD lane with an E-score above 0.9 and an R-risk below 0.13." This is precisely the kind of targeted search that empowers agents to find high-quality, vetted evidence for their tasks. Finally, the `knowledge_eco_score` table introduces a meta-layer of governance, serving as a machine-readable ledger of the eco-impact and risk associated with various entities in the constellation [75]. It allows for KER scores to be attached to repositories, files, schemas, or even documents, providing a quantitative measure of their contribution to the overall mission [95]. An agent could query this table to identify the repositories or shard types with the highest eco-impact and lowest risk, biasing its choices toward the most beneficial and safest options. The complete schema is summarized in the table below.

Table Name	Description	Key Columns
`repo`	Catalog of all repositories in the constellation.	`repo_id`, `name`, `github_slug`, `visibility`, `role_band`
`repo_file`	Index of files within repositories.	`file_id`, `repo_id`, `rel_path`, `ext`, `dir_class`
`aln_schema`	Registry of ALN schema definitions.	`schema_id`, `schema_name`, `version_tag`, `category`
`aln_particle`	Index of ALN particle definitions within schemas.	`particle_id`, `schema_id`, `particle_name`, `role`, `lyap_channel`
`aln_field`	Index of fields within ALN particles.	`field_id`, `particle_id`, `field_name`, `is_risk_coord`, `units`
`corridor_definition`	Registry of risk coordinate safety bands.	`corridor_id`, `varid`, `safe`, `gold`, `hard`, `weight`, `lyap_channel`
`shard_instance`	Registry of concrete evidence shard instances.	`shard_id`, `node_id`, `region`, `lane`, `k_metric`, `r_metric`, `kerdeployable`
`knowledge_eco_score`	Meta-table for KER scores of constellations' objects.	`score_id`, `scope_type`, `scope_ref_id`, `k_factor`, `e_factor`, `r_factor`

The true power of this discovery spine lies in its ability to guide AI agents. By populating this SQLite database, an agent booting up in a new repository can execute a simple SQL

query against the `repo` and `repo_file` tables to get a complete map of the constellation's contents. It can then use the `aln_particle` and `aln_field` tables to understand the meaning of different shard types. When tasked with finding a specific function, the agent can query the `shard_instance` table for shards that meet its criteria for KER scores and deployment lane, ensuring it only uses vetted evidence. The creation of this discovery spine is a pivotal moment, shifting the project from a collection of well-designed individual components to a truly integrated, self-governing system where autonomous agents can operate safely and effectively.

Master-Index Shards: Self-Describing Safety Contracts for Repositories

The concept of a "master-index SQL-shard" per repository is a brilliant piece of engineering that embodies the principle of self-documentation. It acts as a portable, machine-readable "contract" that each repository signs, explicitly declaring its purpose, capabilities, and constraints. This shard is the Rosetta Stone of the EcoNet constellation, decoupling the physical location of a repository from its logical role and capabilities. By embedding this information directly into the repository as a verifiable artifact, the design ensures that governance is portable, auditable, and automatically discoverable by any compliant agent, significantly reducing the risk of misconfiguration and misuse [99] [148]. The shard is designed to answer an agent's questions before it writes a single line of code, transforming the repository from a black box into an open book. The proposed shard contains two main tables: `econet_repo_index` and `econet_layer`. The `econet_repo_index` table holds essential metadata about the repository as a whole, providing a macro view of its role in the larger constellation. Its fields are carefully chosen to capture the most critical governance parameters. Key columns include `repo_name`, `github_slug`, and `role_band` (e.g., 'ENGINE', 'MATERIAL'), which immediately tells an agent the repository's strategic purpose and the set of rules it must follow [29]. Other crucial fields are `ecosafety_binding`, which specifies the exact version of the frozen ecosafety core grammar the repo must use (e.g., 'cyboquatic-ecosafety-core=EcosafetyGrammar2026v1.aln'), and `shard_protocol`, which defines the schema standard it adheres to (e.g., 'ALN-RFC4180/EcoNetSchemaShard2026v1') . To guide deployment, the shard includes `lane_default` (the default operational lane, e.g., 'PROD') and a set of `ker_target_*` values (e.g., `ker_target_k=0.94`, `ker_target_e=0.90`, `ker_target_r=0.12`), which represent the minimum acceptable performance for production artifacts from that repository . Finally, a boolean

`non_actuating_only` flag indicates whether the repository is strictly for analysis and cannot contain any actuator-related code, a critical safety constraint .

While `econet_repo_index` provides the "who" and "why," the `econet_layer` table provides the "how." It details the internal architecture of the repository, specifying the different "layers" it contains. This design elegantly accommodates the requirement to allow a single repository to house multiple, compatible layers while still being bound to a single `role_band` [117][148]. For example, an `ENGINE`-band repository like `EcoNet-CEIM-PhoenixWater` can have multiple entries in `econet_layer`: one for its `GRAMMAR_CLIENT` layer (which imports the core ecosafety types), another for its `KERNEL` layer (which contains the actual CEIM/CPVM computation logic), and perhaps a third for an `EDGE_SCRIPT` layer (which might handle local sensor data). Each layer entry specifies its `layer_name` (e.g., 'KERNEL', 'MATERIAL_MODEL'), the `languages` it uses (e.g., 'Rust,C'), and a `contracts` field that lists the ALN traits or Rust traits it must satisfy (e.g., 'SafeKernel', 'NonActuatingWorkload') . This provides a fine-grained view of the repository's internal composition and allows an agent to find the exact piece of functionality it needs. The shard also includes an optional `econet_role_hint` table to further assist agents. This table allows repository maintainers to attach free-form key-value pairs that describe the repository's domain focus. For example, hints like ('pilot_domain', 'Phoenix water remediation') or ('primary_plane', 'hydraulics') can help an agent rank and filter repositories during a search, moving beyond rigid role-based matching to more contextual relevance [148].

The design explicitly enforces the "one role-band-per-repo" rule, which is a critical enforcement mechanism for the entire governance model. The invariant is simple: there is exactly one row in `econet_repo_index` for a given repository, and that row has a single `role_band` value. All entries in the `econet_layer` table for that repository must be consistent with this band's defined layer set. For instance, if a repository's index declares a `role_band` of 'RESEARCH', no corresponding layer can claim to be a 'CONTROLLER' that interacts with hardware. This prevents a repository from becoming a "jack of all trades" and potentially violating governance rules. Any violation of this contract can be caught by a CI linting job that validates the shard against a master policy document, turning the manifest from a suggestion into a hard requirement. This ensures a clear audit trail and simplifies automated verification, reinforcing the integrity of the entire system. Below is a template for the master-index shard, showing how it would be populated for two different repositories, `EcoNet-CEIM-PhoenixWater` and `BugsLife`.

```
-- .econet/econet_repo_index.sql
PRAGMA foreign_keys = ON;
```

```
CREATE TABLE IF NOT EXISTS econet_repo_index (
    repo_name TEXT PRIMARY KEY,
    github_slug TEXT NOT NULL,
    role_band TEXT NOT NULL, -- 'SPINE','RESEARCH','ENGINE','MATERIAL','GO
    visibility TEXT NOT NULL, -- 'Public','Private'
    language_primary TEXT NOT NULL,
    description TEXT,
    ecosafety_binding TEXT NOT NULL,
    shard_protocol TEXT NOT NULL,
    lane_default TEXT NOT NULL,
    ker_target_k REAL NOT NULL,
    ker_target_e REAL NOT NULL,
    ker_target_r REAL NOT NULL,
    non_actuating_only INTEGER NOT NULL CHECK (non_actuating_only IN (0,1)
);

CREATE TABLE IF NOT EXISTS econet_layer (
    layer_id INTEGER PRIMARY KEY AUTOINCREMENT,
    repo_name TEXT NOT NULL REFERENCES econet_repo_index(repo_name) ON DEL
    layer_name TEXT NOT NULL,
    layer_tier TEXT NOT NULL, -- 'GRAMMAR','KERNEL','EDGE_SCRIPT','UI','GO
    languages TEXT NOT NULL,
    description TEXT,
    contracts TEXT
);

INSERT OR REPLACE INTO econet_repo_index VALUES
(
  'EcoNet-CEIM-PhoenixWater',
  'Doctor0Evil/EcoNet-CEIM-PhoenixWater',
  'ENGINE',
  'Public',
  'Rust',
  'Phoenix water CEIM/CPVM kernels and ecosafety-controlled controllers.',
  'cyboquatic-ecosafety-core=EcosafetyGrammar2026v1.aln',
  'ALN-RFC4180/EcoNetSchemaShard2026v1',
  'PROD',
  0.94, 0.90, 0.12,
  0
```

```
);

INSERT INTO econet_layer VALUES
('EcoNet-CEIM-PhoenixWater', 1, 'Ecosafety spine client', 'GRAMMAR', 'Rust
('EcoNet-CEIM-PhoenixWater', 2, 'CEIM/CPVM hydrology kernels', 'KERNEL', '
('EcoNet-CEIM-PhoenixWater', 3, 'Hydrological buffer atlas', 'KERNEL', 'Ru
```

This master-index shard is the definitive source of truth for intra-constellation communication and collaboration. It is a lightweight yet powerful mechanism for embedding governance directly into the assets it governs, ensuring that the complex rules of the EcoNet constellation are always accessible, understandable, and enforceable by both humans and machines.

Implementing Governance: From Repository Roles to Cross-Plane Invariants

The true power of the EcoNet architecture emerges when the foundational pillars—the frozen grammar, the discovery spine, and the master-index shards—are woven together to create a cohesive governance fabric. This integration transforms the system from a static collection of rules into a dynamic, self-enforcing ecosystem. The implementation of governance hinges on codifying abstract concepts into concrete, verifiable artifacts and integrating them into the CI/CD pipeline. This involves creating new Rust traits to formally define safe workloads, finalizing multi-plane invariants to prevent harmful trade-offs, and introducing novel risk metrics to monitor the health of the constellation itself. Each of these steps strengthens the safety contract, raises the Knowledge-factor (K), and increases the Eco-impact (E) while keeping the Risk-of-harm (R) in a bounded, low range.

A primary area for formalization is the concept of a "non-actuating workload," which has been used operationally but lacks a formal definition in the codebase [148]. The first step is to define a `trait NonActuatingWorkload` in the ecosafety spine. This trait would serve as a compile-time guarantee of safety, ensuring that any crate implementing it adheres to a strict set of rules. A potential signature could be:

```
/// A trait for workloads that compute ecosafety metrics from telemetry wi
pub trait NonActuatingWorkload {
    /// Inputs are typically snapshots of telemetry or data shards.
```

```
    type Input;
    /// Outputs include the updated risk vector, Lyapunov residual, and KE
    type Output;

    /// Executes the workload, transforming inputs into outputs.
    /// This function must not, under any circumstances, directly or indir
    /// call functions that control physical actuators (e.g., PLCs, motors
    fn execute(&self, input: Self::Input) -> Self::Output;
}
```

By having all existing non-actuating crates—such as `EnergyToMassKernels`, `MaterialKinetics`, and diagnostic frames like `HydraulicDecayFrame`—implement this trait, the discovery spine gains a reliable way to tag which functions are safe for agents to call [23] [35]. This moves beyond heuristic-based classification (e.g., inferring safety from a repository's name) to a rigorous, machine-verifiable standard. Unit tests for each implementation would be required to prove monotonicity (that increasing harmful inputs lead to higher or equal risk coordinates) and boundedness (that all outputs remain within the `[0,1]` range), providing mathematical proof of correctness.

Another critical aspect of governance is the prevention of "greenwashing" or harmful trade-offs between different ecological planes. The system already recognizes the need for non-compensation invariants, particularly for sensitive planes like `carbon` and `biodiversity` [36]. The next step is to formalize this into a concrete ALN specification, tentatively named `PlaneWeightsShard2026v1.aln`. This shard would define not only the weights for each plane but also flags indicating which planes are non-offsettable. For example, the shard might contain rows like: `varid: 'CARBON_NONOFFSETTABLE', value: true, mandatory: true`. With this data available in the discovery spine's `corridor_definition` table, a governance shard or a CI invariant can enforce a specific rule: if a risk coordinate for a non-offsettable plane rises above a certain threshold (e.g., the 'gold' band) and worsens over a timestep, then the `R` metric must not decrease and the `E` metric must not increase. This can be encoded as a simple predicate on consecutive KER windows, mathematically forbidding an agent or optimizer from finding a solution that improves energy efficiency at the cost of increased carbon emissions or habitat loss [56]. This prevents the system from converging on superficially optimal but ecologically catastrophic solutions.

Furthermore, the governance layer can be extended to monitor the health of the constellation itself. A novel and powerful idea is to introduce a new risk coordinate, `rtopology`, which quantifies the degree of misalignment between a repository's declared manifest and its actual contents [90]. This would involve creating a CI job that

performs a deep check of every repository. The raw score, `I_topology`, would be calculated as a weighted sum of errors detected over a window of time: $I_{topology} = w_{mislabel} N_{mislabel} + w_{missing} N_{missing}$, where $N_{mislabel}$ is the count of repos whose actual traits conflict with their declared `role_band`, and $N_{missing}$ is the count of repos that exist in the organization but lack a `.econet/econet_repo_index.sql` shard entirely. The weights emphasize mislabels over missing entries (w_{mislabel} \ge w_{missing} > 0). This raw score is then normalized using the standard piecewise linear corridor kernel used for other risk coordinates to produce a value for `rtopology` in the range $[0,1]$ [18]. This new risk coordinate is then added to the Lyapunov residual, modifying it to $V_t' = V_t + w_{topology} r_{topology}^2$, with $w_{topology} \geq 0$. Since this is a non-negative term, all existing proofs about the properties of the Lyapunov function remain valid; however, the admissible region of the system shrinks or stays the same, never expands. This elegant mechanism makes governance errors themselves a measurable form of ecological risk, creating a powerful feedback loop. As the constellation grows, a rising `rtopology` will push up the overall residual, making it harder for the system to achieve a deployable state until the underlying governance issues are corrected. This turns the maintenance of the constellation's integrity into a direct optimization problem. The following table summarizes these governance enhancements.

Artifact / Concept	Description	KER Impact
`NonActuatingWorkload` Trait	A Rust trait that formally defines and enforces the contract for non-actuating computations.	**K↑** : Provides a compile-time, machine-verifiable guarantee of safety, eliminating ambiguity and heuristics.
`PlaneWeightsShard2026v1.aln`	An ALN shard defining per-plane weights, non-compensation flags, and uncertainty caps.	**K↑** : Hardcodes critical invariants (e.g., non-offsettable planes) into the grammar, preventing silent weakening.
`rtopology` Risk Coordinate	A new risk coordinate measuring the alignment between repository manifests and their actual contents.	**K↑** : Creates a direct incentive to maintain accurate and honest repository manifests, improving the fidelity of the discovery spine.
CI Linting for Repo Indexes	A CI job that validates every `.econet/econet_repo_index.sql` file against a master policy.	**K↑** : Ensures uniform application and accuracy of governance manifests across the entire constellation.

By systematically implementing these governance mechanisms, the EcoNet constellation evolves from a promising design into a robust, industrial-grade platform. The KER scores for this work remain firmly in the desired range: K ≈ 0.95 (due to the formalization of safety contracts and topology monitoring), E ≈ 0.92 (as the system biases itself towards properly governed, eco-restorative functions), and R ≈ 0.12 (as all new artifacts are descriptive or non-actuating, and existing safety invariants are only tightened).

The Role of Virta-Sys: Enforcing Invariants Before Optimization

The user's clarification on Virta-Sys's role is paramount. It is not merely an orchestrator but a "governance-first headquarters." Its authority flows from enforcing the frozen ecosafety contract before any other consideration. This principle dictates that Virta-Sys must treat governance enforcement as the primary function, with energy-cost reduction and cross-band coordination built on top of that governance core rather than competing with it. The constellation's safety model is frozen around Lyapunov residuals, K/E/R thresholds, lanes, and non-offsettable planes; if Virta-Sys ever optimizes energy or coordination without first enforcing those invariants, it would undermine the very reason it exists. The ordering of operations is therefore: (a) enforce ecosafety and lane rules, (b) ensure all bands are aligned on the same frozen grammar and corridors, and then (c) within that safe envelope, push hard on energy-cost reduction and placement optimization. This positioning establishes Virta-Sys as the authoritative decision engine for several critical functions, acting as a non-actuating service that reads the frozen grammar and constellation index, computes decisions, and writes verdicts (ALN/SQL shards), leaving the final scheduling and actuation to other repositories. However, it holds strong advisory authority plus hard veto power on governance invariants, making it the decisive gatekeeper for what can and cannot run where [124].

The core functionality and authority of Virta-Sys are realized through a suite of integrated services. As the **Lane & Tier Governor**, it acts as the arbiter of the RESEARCH → EXPPROD → PROD transition ladder. It implements a formal lane transition function that maps (kernel, region, evidence history) to a lane, with explicit preconditions based on K/E/R metrics and adherence to safety corridors [101]. This is implemented through a `virta_lane_verdict` table that other repos, such as `ecological-orchestrator` and `eco_infra-governance`, must query and respect. Any attempt to promote a workload to a higher lane without a corresponding "Admissible" verdict from Virta-Sys must be treated as a hard block by CI and runtime pipelines [148]. As the **Constellation Topology Auditor**, it continuously checks repo manifests (`.econet` shards), role_bands, and layer_tiers against actual code footprints, feeding the `rtopology` risk coordinate into the Lyapunov residual [90]. This turns governance drift into a quantifiable risk that impacts the system's deployability [69]. As the **Evidence-Aware Scheduler** and **Energy-Cost Cartographer**, it operates within the safe envelope defined by the ecosafety invariants. It recommends kernel placements that minimize energy cost per unit of KER improvement. This is achieved by computing a cost-efficiency metric, $J(W,n)=E_{node}(W)/\Delta E(W)$, subject to the constraint that non-offsettable planes are

never compromised [58] [77]. This turns optimization into a constrained problem inside a safe ecological corridor [32]. As the **Cross-Band Coordination Hub**, it mediates between ENGINE, MATERIAL, GOV, and APP bands, ensuring that controllers, materials, finance/reward flows, and apps all reference the same frozen grammar, corridor definitions, and KER thresholds when they coordinate actions [96].

The authority of Virta-Sys is codified through concrete research objects and upgrade pathways. First, **freezing lane governance** requires defining an ALN shard and SQL tables (`virta_lane_verdict`) for lane transitions. A small Rust service in Virta-Sys would compute verdicts based on `shard_instance` history, and CI in `eco_infra-governance` would be wired to treat missing/Denied verdicts as hard blocks [128][130]. Second, **implementing the placement cost metric** involves formalizing the `J(W,n)` formula and adding a `virta_placement_verdict` table. This allows `ecological-orchestrator` to consult these verdicts before scheduling workloads onto nodes [63] [181]. Third, **materializing `rtopology`** requires implementing the topology audit job in `eco_infra-governance` (or within Virta-Sys). A `TOPOLOGY.MISALIGNMENT` row is added to `corridor_definition`, and `rtopology` is incorporated into the global Lyapunov residual calculation, making governance drift a first-class eco risk [131][132]. Fourth, **locking in `NonActuatingWorkload`** involves adding the trait to `EcoNet` and having key repos implement it. A `virta_contract_registry` table and CI checks verify implementations, ensuring claims of being non-actuating are backed by code [35] [148]. Finally, **introducing a sovereign improvement ledger** requires implementing `virta_upgrade_ledger` to track changes in K/E/R for schemas and kernels. Requiring ledger entries for significant changes, enforced by CI, ensures monotonic improvements and provides agents with a history of proven performance [135]. The table below outlines the roles and their corresponding research objects.

Virta-Sys Role	Core Functionality	Key Research Object	Governance Authority Mechanism
Lane & Tier Governor	Acts as the arbiter of RESEARCH → EXPPROD → PROD transitions.	`virta_lane_verdict` SQL table and ALN shard.	Hard veto power via CI checks on `verdict` status.
Constellation Topology Auditor	Quantifies governance drift and feeds `rtopology` into the Lyapunov residual.	`rtopology` risk coordinate and associated audit tables.	Raises the global Lyapunov residual, making unsafe states harder to reach.
Energy-Cost Cartographer	Computes cost-efficiency metrics (`J(W, n)`) for kernel placement.	`virta_placement_verdict` SQL table with `cost_per_delta_E` field.	Strong advisory authority; `ecological-orchestrator` consults verdicts.
Cross-Band Coordination Hub	Ensures all bands use the same frozen grammar and corridor definitions.	`workflow` table and `band_alignment_checker` module.	Rejects workflows with mismatched grammar/corridor versions.
Experiment Routing Steward	Manages EXPPROD experiments and decides on promotion to PROD.	`experiment` and `experiment_window` SQL tables.	Issues `ExperimentPromotionShard` upon meeting performance targets.
KER Performance Archivist	Maintains long-horizon histories of K/E/R per workload and region.	`workload_ker_summary` materialized view/table.	Provides trusted data for `ecological-orchestrator` to select high-performing workloads.

By pursuing these research objectives, the EcoNet constellation moves from a promising design to a robust, industrial-grade platform where safety is a structural property, not an afterthought. Virta-Sys, as the governance-first headquarters, becomes the indispensable brain of the constellation, ensuring that all growth and optimization are additive and safe.

Synthesis and Strategic Pathway: Formalizing the Constellation's Governing Logic

In synthesizing the preceding analysis, a clear and actionable pathway emerges for realizing the full potential of the EcoNet constellation. The proposed research framework successfully translates high-level ecological goals into a low-level, computationally tractable contract, making it a robust model for governing complex, distributed systems with high-stakes outcomes. The framework's success is predicated on its four foundational pillars: a frozen ecosafety grammar enforced by spine repositories; a strict spine/role-band architecture; a centralized, queryable discovery spine (SQLite database); and self-describing master-index shards for each repository. The role of Virta-Sys is then defined as the non-actuating "brain" that enforces these invariants through a suite of integrated services, including lane governance, topology auditing, and energy-aware

placement, all while acting as a queryable service for the entire constellation. The synthesis reveals that the most impactful next steps involve crystallizing the conceptual roles of Virta-Sys into concrete, testable artifacts that can be integrated into the CI/CD pipeline, thereby elevating the system from a promising design to a verified, industrial-grade platform.

The strategic pathway forward is to pursue a series of concrete research objectives that formalize the governance logic. First, **freezing lane governance** requires defining an ALN shard and SQL tables (`virta_lane_verdict`) for lane transitions. A small Rust service in Virta-Sys would compute verdicts based on `shard_instance` history, and CI in `eco_infra-governance` would be wired to treat missing/Denied verdicts as hard blocks [128][130]. This solidifies Virta-Sys's authority as the gatekeeper for production readiness. Second, **implementing the placement cost metric** involves formalizing the `J(W,n)` formula and adding a `virta_placement_verdict` table. This allows `ecological-orchestrator` to consult these verdicts before scheduling workloads onto nodes, ensuring that optimization occurs within the safe envelope [63] [181]. Third, **materializing `rtopology`** requires implementing the topology audit job in `eco_infra-governance` (or within Virta-Sys). A `TOPOLOGY.MISALIGNMENT` row is added to `corridor_definition`, and `rtopology` is incorporated into the global Lyapunov residual calculation, making governance drift a first-class eco risk that directly impacts system stability [131][132]. Fourth, **locking in `NonActuatingWorkload`** involves adding the trait to `EcoNet` and having key repos implement it. A `virta_contract_registry` table and CI checks verify implementations, ensuring that claims of being non-actuating are backed by code, dramatically raising the Knowledge factor (K) and reducing the Risk-of-harm (R) [35] [148].

Finally, **introducing a sovereign improvement ledger** requires implementing `virta_upgrade_ledger` to track changes in K/E/R for schemas and kernels. Requiring ledger entries for significant changes, enforced by CI, ensures monotonic improvements and provides agents with a history of proven performance [135]. These interconnected upgrades collectively transform the EcoNet constellation into a self-governing system. The discovery spine becomes the ultimate authority, and any code generation tool is merely a sophisticated client of that authority. The risk is kept low because every generation step starts from a ground truth of hex-stamped evidence, preventing the introduction of novel, unvetted logic that could undermine the frozen ecosafety contract [95]. The result is a system where autonomous agents can operate safely and effectively, guided by a verifiable, semantically rich environment where ecological harm is made structurally impossible.

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